NaturArv3D and Compression Strategies

Qim Working Group

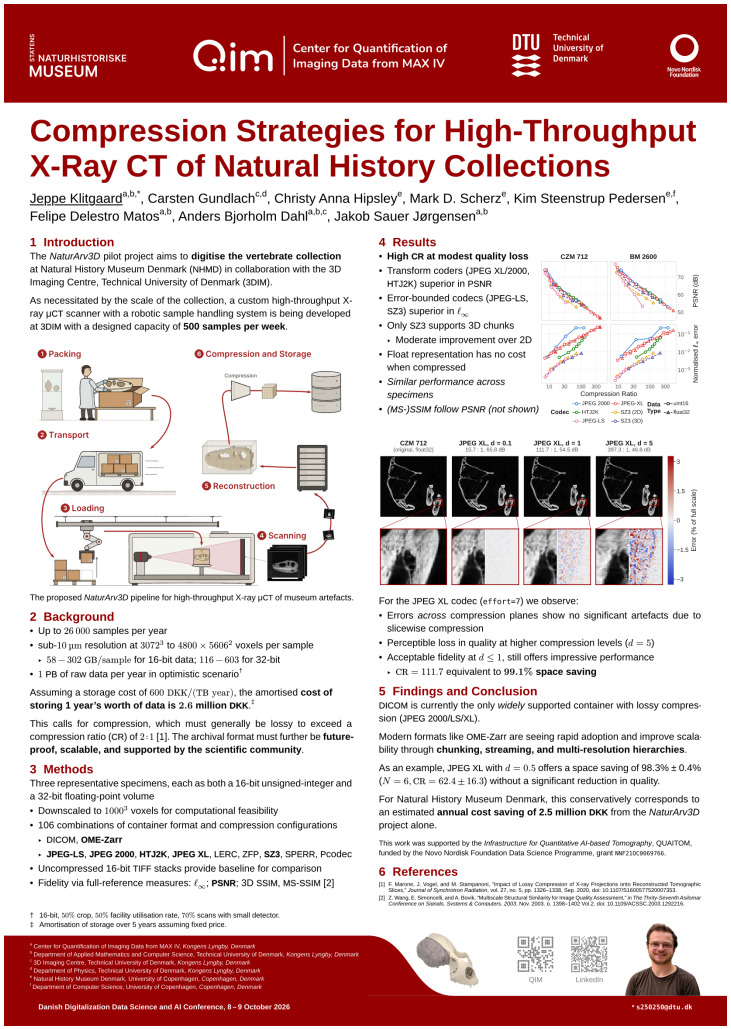
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Jeppe Klitgaard

jepkl@dtu.dk

Poster at D3A 4.0

- Presented this work at D3A 4.0 conference recently
  - Compression of XCT volumes
  - Within context of NaturArv3D project



NATURHISTORISKE MUSEUM | Center for Quantification of Imaging Data from MAX IV | DTU Technical University of Denmark | Novo Nordisk Foundation

Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

Jeppe Klitgaard[a,b,*], Carsten Gundlach[c,d], Christy Anna Hipsley[e], Mark D. Scherz[e], Kim Steenstrup Pedersen[e,f], Felipe Delestro Matos[a,b], Anders Bjorholm Dahl[a,b,c], Jakob Sauer Jørgensen[a,b]

1 Introduction

The *NaturArv3D* pilot project aims to **digitise the vertebrate collection** at Natural History Museum Denmark (NHMD) in collaboration with the 3D Imaging Centre, Technical University of Denmark (3DIM).

As necessitated by the scale of the collection, a custom high-throughput X-ray µCT scanner with a robotic sample handling system is being developed at 3DIM with a designed capacity of **500 samples per week**.

The proposed *NaturArv3D* pipeline for high-throughput X-ray µCT of museum artefacts.

2 Background

- Up to $26\,000$ samples per year
- sub-$10\,\mu\mathrm{m}$ resolution at 3072^3 to 4800×5606^2 voxels per sample
  - $58 - 302\ \mathrm{GB/sample}$ for 16-bit data; $116 - 603$ for 32-bit
- $1\ \mathrm{PB}$ of raw data per year in optimistic scenario†

Assuming a storage cost of $600\ \mathrm{DKK/(TB\ year)}$, the amortised **cost of storing 1 year's worth of data is 2.6 million DKK**.‡

This calls for compression, which must generally be lossy to exceed a compression ratio (CR) of $2{:}1$ [1]. The archival format must further be **future-proof, scalable, and supported by the scientific community**.

3 Methods

Three representative specimens, each as both a 16-bit unsigned-integer and a 32-bit floating-point volume

- Downscaled to 1000^3 voxels for computational feasibility
- 106 combinations of container format and compression configurations
  - DICOM, **OME-Zarr**
  - **JPEG-LS**, **JPEG 2000**, **HTJ2K**, **JPEG XL**, LERC, ZFP, **SZ3**, SPERR, Pcodec
- Uncompressed 16-bit TIFF stacks provide baseline for comparison
- Fidelity via full-reference measures: ℓ_∞; **PSNR**; 3D SSIM, MS-SSIM [2]

† 16-bit, 50% crop, 50% facility utilisation rate, 70% scans with small detector.
‡ Amortisation of storage over 5 years assuming fixed price.

4 Results

- **High CR at modest quality loss**
- Transform coders (JPEG XL/2000, HTJ2K) superior in PSNR
- Error-bounded codecs (JPEG-LS, SZ3) superior in ℓ_∞
- Only SZ3 supports 3D chunks
  - Moderate improvement over 2D
- Float representation has no cost when compressed
- *Similar performance across specimens*
- *(MS-)SSIM follow PSNR (not shown)*

For the JPEG XL codec (`effort=7`) we observe:

- Errors *across* compression planes show no significant artefacts due to slicewise compression
- Perceptible loss in quality at higher compression levels ($d = 5$)
- Acceptable fidelity at $d \leq 1$, still offers impressive performance
  - $\mathrm{CR} = 111.7$ equivalent to $\mathbf{99.1\%}$ **space saving**

5 Findings and Conclusion

DICOM is currently the only *widely* supported container with lossy compression (JPEG 2000/LS/XL).

Modern formats like OME-Zarr are seeing rapid adoption and improve scalability through **chunking, streaming, and multi-resolution hierarchies**.

As an example, JPEG XL with $d = 0.5$ offers a space saving of $98.3\% \pm 0.4\%$ ($N = 6, \mathrm{CR} = 62.4 \pm 16.3$) without a significant reduction in quality.

For Natural History Museum Denmark, this conservatively corresponds to an estimated **annual cost saving of 2.5 million DKK** from the *NaturArv3D* project alone.

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6 References

[1] F. Mancini, J. Vogel, and M. Stampanoni, "Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices," *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007351.
[2] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale Structural Similarity for Image Quality Assessment," in *The Thirty-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Nov. 2003, p. 1398–1402 Vol.2, doi: 10.1109/ACSSC.2003.1292216.

* s250250@dtu.dk

QIM LinkedIn

Danish Digitalization Data Science and AI Conference, 8 – 9 October 2026

NaturArv3D

Joint project between Natural History Museum Denmark (NHMD) and 3D Imaging Centre (3DIM)

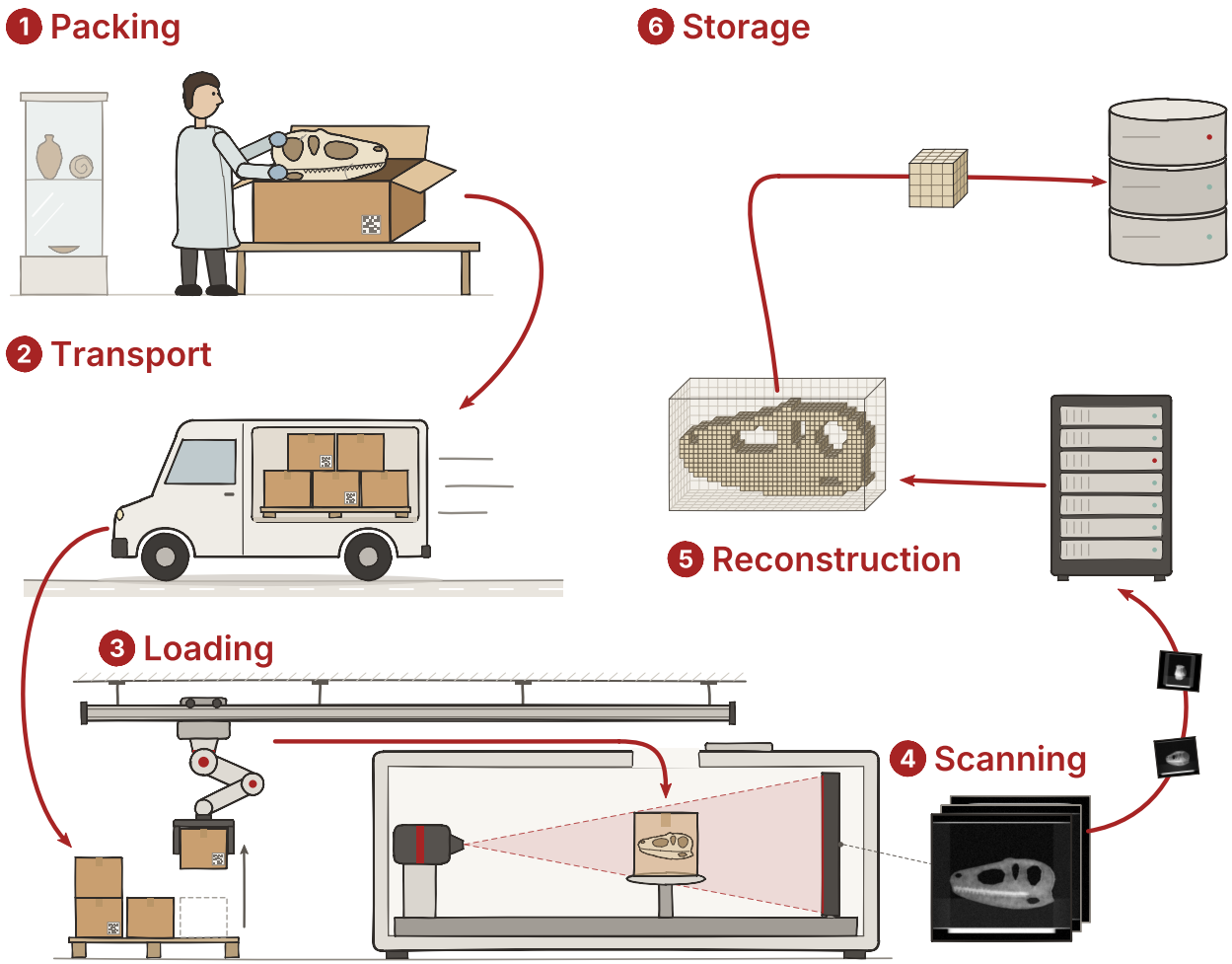
New X-ray scanner and infrastructure:

- **High throughput**: 500 samples/week (≈15 min per sample)
- **High resolution**: resolution below 10 µm, 3K – 5.6K images
- Samples prepared in cardboard boxes
- **Autonomous sample handling** using robot mounted in ceiling gantry

Purpose: Digitize large collections

- Pilot: Seal skulls, vertebrate collection

NaturArv3D


1 Packing
2 Transport
3 Loading
4 Scanning
5 Reconstruction
6 Storage

Data Size Matters

Problem: Data is (very) large!

- Projections: not sure, likely fairly undersampled (15 minutes per sample!)
  - Most people wouldn't save these anyway
- Small detector: 3072×3072 pixels
  - Reconstruction, 32-bit float: 116 GB per sample
- Big detector: 5606×4800 pixels
  - Reconstruction, 32-bit float: 603 GB per sample

Assume 50/50% small/big detector usage, 500 samples/week:

- Recons: 9384 TB per year (!!)

Assume storage cost of 600 DKK / (TB yr)

- Assume data must be stored for 5 years: 28.2 MDKK / yr

🌀 *Money clearly better spent giving Jeppe salary bump* 🌀

Low-hanging fruit

1. 50/50% mix, 500 samples/week, f32, no crop:
   - 9384 TB / yr (28.2 MDKK / yr)
2. 16-bit representation:
   - 4692 TB / yr (14.1 MDKK / yr)
3. Crop 50% of volume:
   - 2346 TB / yr (7 MDKK / yr)
4. Maybe actually scanner utilisation is only 250 samples / week (50%):
   - 1173 TB / yr (3.5 MDKK / yr)

1 minute and 43 seconds of Anthropic net loss 2026-H1

That is really not that much money!

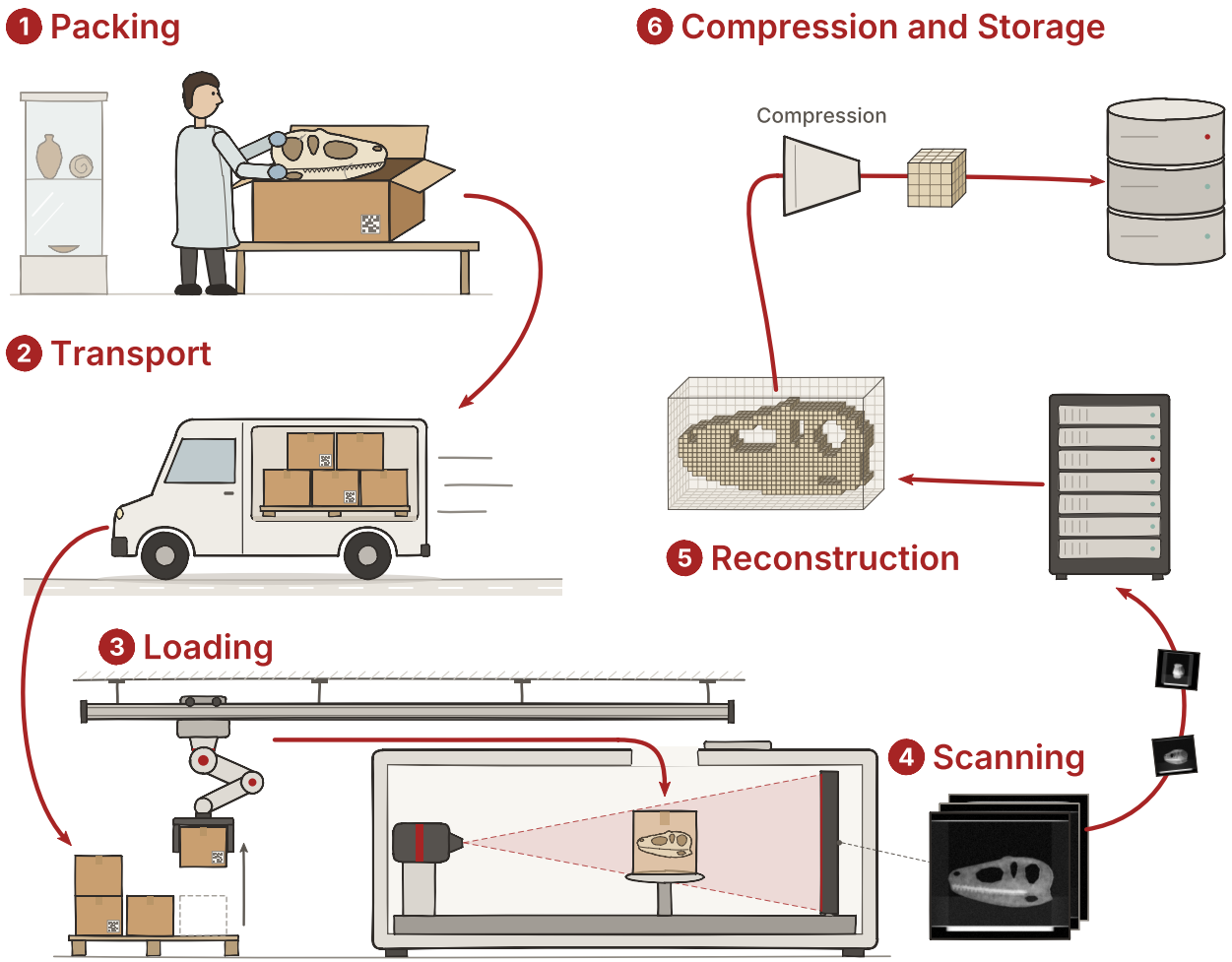
Going Further

How to get around this?

- Reduce bit depth further: 16-bit ⟶ 8-bit
  - Saves 50% of cost, but **99.6% reduction in bit depth**!
- Downsample images?
- Crop more?
- Use big detector less?
- Just scan fewer samples?

 ***What about compression?*** 

Compression



Compression

- Criteria
  - Ideally lossless, otherwise as little quality loss as possible
  - Decompression speed must not be bottleneck
  - Format must be suitable for archival and research
    - Wide software support
    - Format stable over time
  - Format should support metadata (voxel size, etc)
  - Worthwhile compression ratio
- Dataset
  - 3 volumes of seal skulls, downsampled to 1K
  - Both `uint16` and `float32` versions tried

Formats

Format	ImageJ/Fiji	Napari	VGSTUDIO	Amira/Avizo	3DSlicer	Dragonfly	Checkpoint	Morphosource	Data Types	Comment
BioFormats	Yes[1]	Yes[1]	No	Yes	No	No	No	No		
TIFF	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u/i[8/16/32], f16/32	
BigTiff	Yes	Yes[1]	No?	Yes	No?	No?	No?	No?	u/i[8/16/32], f16/32	
JPEG-1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u8 (u16 LL)	
JPEG-2000	Yes	?	Yes	Yes	No?	No?	No?	Yes?	u1-38	
JPEG-XR	No[2]	?	No?	No[2]	No?	No?	No?	No?	u8/16/32	
JPEG-LS	No?	?	No?	No?	No?	No?	No?	No?	u8/16/32	
JPEG-XL	No	Yes[1]	No	No	No	No	No	No	u8/16/32	
DICOM	Yes	Yes[1]	Yes	Yes	Yes	Yes	Yes	Yes	u8/16	Likely no JXL support?
OME-XML	Yes[1]	Yes[1]	No	Yes	No	No	No	No	u/i[8/16/32], f16/32, …	
OME-TIFF	Yes[1]	Yes[1]	No	Yes	No?	No?	No?	No	u/i[8/16/32], f16/32, …	
OME-ZARR	Yes[1]	Yes[1]	No	Yes[3]	No	No	No	No	u/i[8/16/32], f16/32, …	
HDF5	Yes	Yes[1]	No[4]	Yes[1]	No	Yes?	No	No	u/i[8/16/32], f16/32, …	
PNG	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes?	u8/16	
PPM	Yes	?	Yes	Yes	No?	No?	No	No	u8/16	
RAW	Yes	?	Yes	Yes	Yes	Yes	No	No?		
BMP	Yes	?	Yes	Yes	Yes	Yes	No	Yes	u8	
Analyze/HDR	Yes?	Yes[1]	No	Yes	Yes	Yes	Yes	No	u8, i16/32, f32/64, …	
NifTI	Yes	Yes[1]	No	Yes	Yes	Yes	Yes	No	u/i[8/16/32], f32/64, …	
Other notable					Many		FITS (.fits), Kitware VTK (.vtk)			
Source			E-Mail with *Technical Solutions Export*		3D Slicer Wiki	Dragonfly Manual	Checkpoint	Duke		
Comment	FOSS (GPLv3)	FOSS (BSD-3), Plugin system	Proprietary	Proprietary	FOSS (BSD), Anything via Python		Proprietary	Museum data repository		

1 With plugin
2 Supported through `.czi` format
3 With plugin by Janelia
4 Does support HDF4

Formats

- TIFF
  - Extremely common, almost always without compression
  - Supports many codecs through *extensions*
    - Almost no software support in practice
- DICOM
  - Widely supported, popular in medical field
  - Natively supports JPEG 2000, HTJ2K, JPEG-LS, (JPEG XL)
  - Cumbersome format to deal with, 3DIM strongly dislikes
- Bitmap stacks
  - Stacks of PNG, JPEG, JPEG 2000, …
  - Either low bit-depth or not widely supported

Formats

- Conclusion:
  - DICOM is the only widely supported format supporting lossy compression!
  - (with > 8 bit depth)
- Generally lossy required to exceed $\mathrm{CR} \approx 2{:}1$ [1] [2]

Leaves us with ~~3~~ **1** option:

1. ~~Wide support + low/no compression: TIFF stacks~~ **Can't afford**
2. ~~Wide support + lossy compression: DICOM~~ **3DIM says no** (cumbersome)
3. Limited support + lossy compression: Free to choose modern format

OME-Zarr

(or other NGFF)

[1] F. Marone, J. Vogel, and M. Stampanoni, "Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices," *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

[2] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and R. Herfkens, "Real-Time Compression of Raw Computed Tomography Data: Technology, Architecture, and Benefits," presented at the SPIE Medical Imaging, E. Samei and J. Hsieh, Eds., Lake Buena Vista, FL, Feb. 2009, p. 72582H. doi: 10.1117/12.810599.

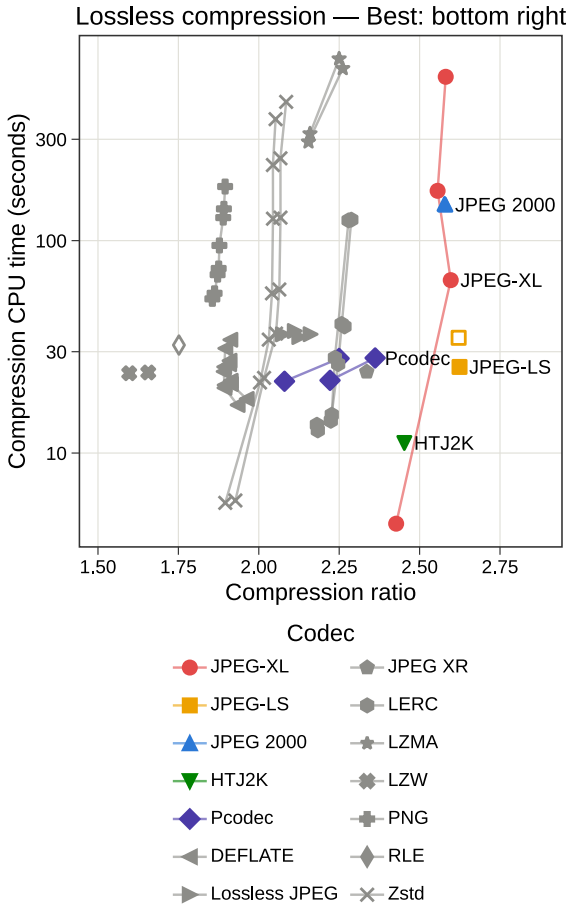
Formats

- OME-Zarr
  - Sparse software support, but rapidly improving (FOSS especially)
  - Excellent support in Python and on Web, (also Rust)
  - Advanced features
    - chunking, sharding
    - multiscale
    - streaming
    - metadata

    } Important for large datasets
  - Comes from microscopy community, some features not relevant

Lossless compression

- 375 combinations of codecs, parameters, dataset
- JPEG codecs perform well, including JPEG XL
  - Very recent format
    - Adopted by browsers in last 2 months
    - Next PDF standard, latest DICOM spec. [1]
- $\mathrm{CR} \approx 2.8{:}1$ best case (JPEG-LS)
- Why do we get $\mathrm{CR} > 2{:}1$, contrary to literature?
  - Wolf skulls reconstructed with CIL compress worse
  - Quirk of the seal skull dataset
    - Masked corners (const value outside reconstruction circle)
    - Lower noise than CIL reconstructions



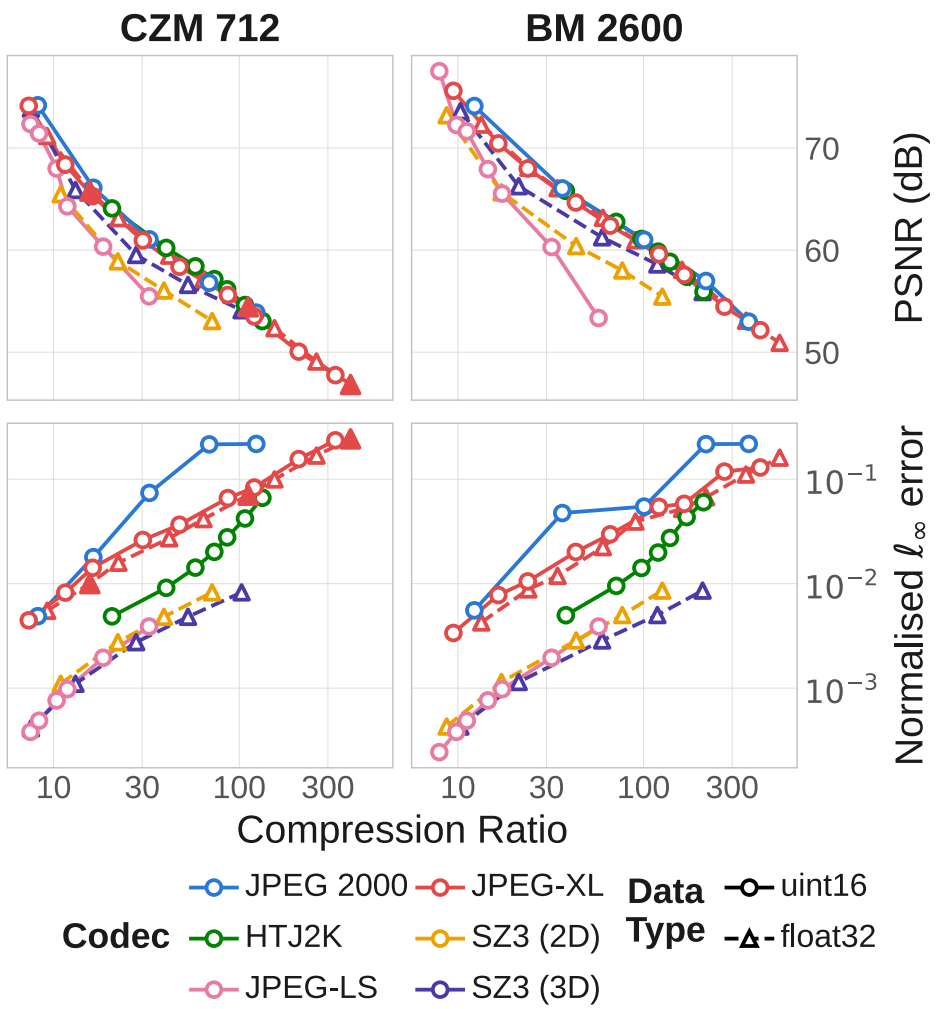
[1] "DICOM PS3.5 2026c — Data Structures and Encoding." Accessed: Jul. 17, 2026. [Online]. Available: https://www.dicomstandard.org/current

Lossy compression

- 318 combinations of codecs, parameters, dataset
- For our 2×3 seal skull datasets try a bunch of different codecs in **DICOM** and **OME-Zarr**
  - JPEG-LS, JPEG 2000, HTJ2K, JPEG XL, LERC, ZFP, SZ3, SPERR
- Compute full-reference measures against original data
  - ℓ_∞, PSNR, 3D SSIM, MS-SSIM

Lossy compression

- **Large CR with relatively modest quality loss!**
- SZ3: $\ell_\infty < 10^{-2}$ gives $\mathrm{CR} > 100$
- Transform coders superior in PSNR
  - JPEG XL, JPEG 2000, HTJ2K
- Error bounded codecs superior in ℓ_∞
  - JPEG-LS, SZ3



Lossy compression

- JPEG XL:
  - `f32` representation is ‘free’ when compressed

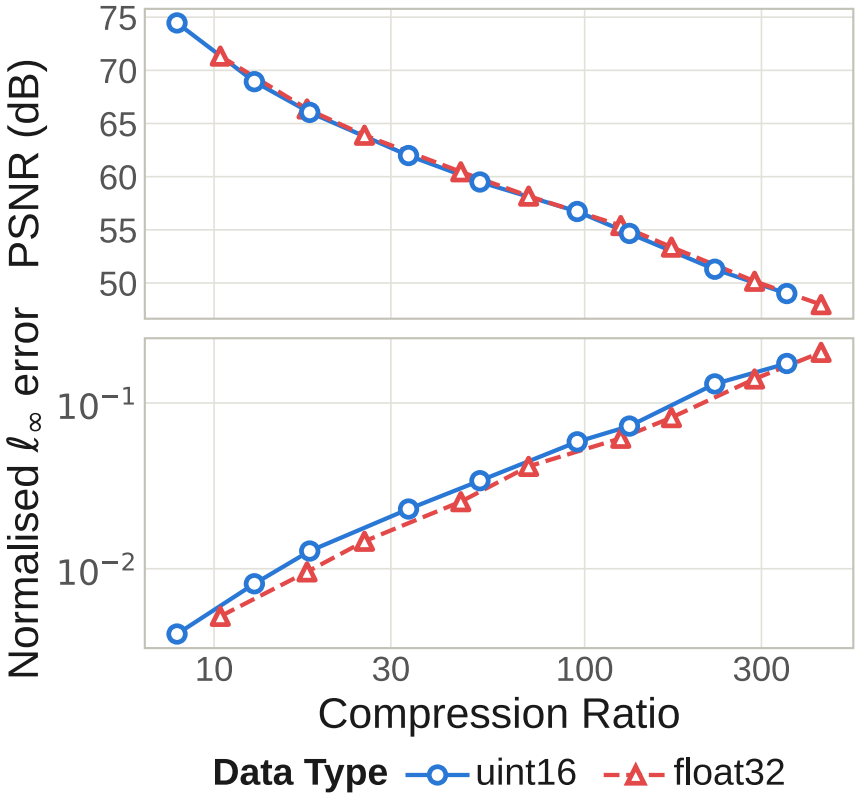


Figure 3: Compression ratio against uint 16-bit baseline for JPEG XL with $d \in [0.05, 5.0]$, `effort=7` across the 3 seal skull datasets.

Lossy compression

- SZ3 2D vs 3D to gauge cost of 2D compression in 3D volumes
- Performs better given 3D data than 2D slices
- Difference small enough that 2D codecs are still viable for 3D volumes

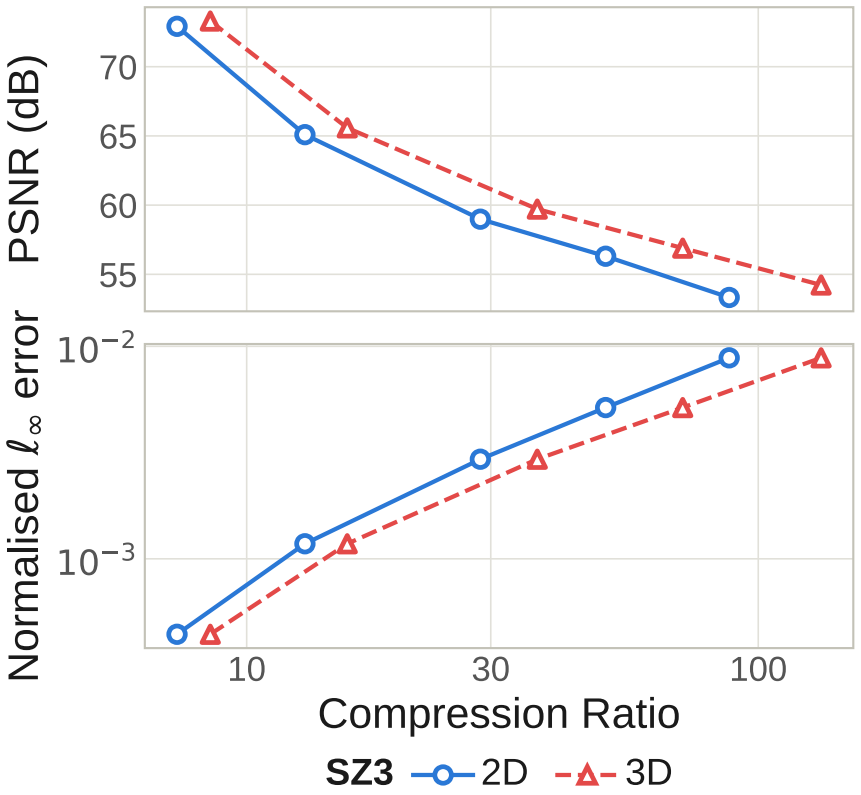


Figure 4: Compression ratio for SZ3 using 3D chunks vs 2D slices across the 3 seal skull datasets.

Lossy compression

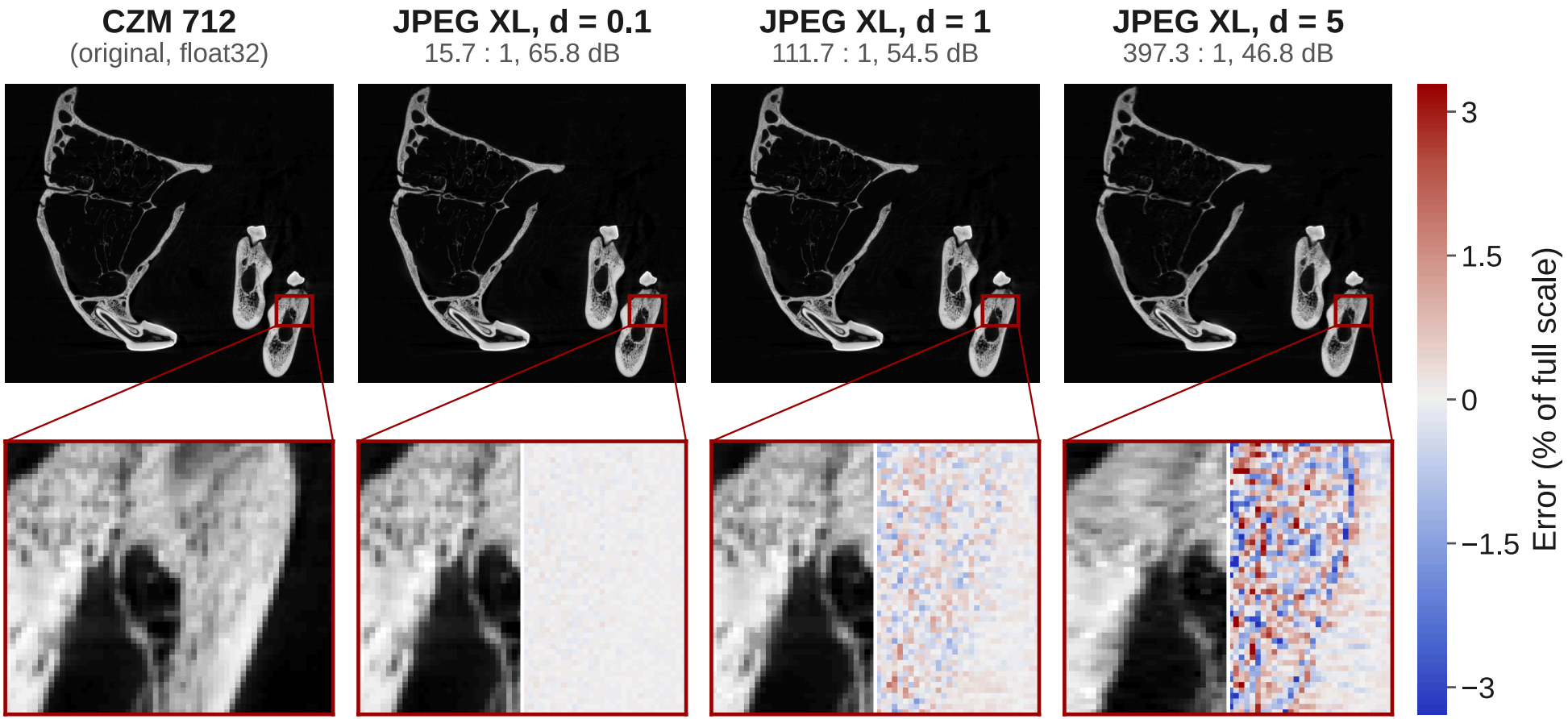


Figure 5: Compression artefacts from JPEG XL (2D) compression of 3D volume (CZM 712) *across* compression slices for different quality settings.

Lossy compression

- As an example, JPEG XL with $d = 0.5$
- No perceptible errors (abundance of caution)
- For dataset $N = 6$ gives $\mathrm{CR} = 62.4 \pm 16.3$
- $\Rightarrow 98.3\% \pm 0.4\%$ **space savings!**

$$3.5\ \mathrm{MDKK} \rightarrow 59\,821\ \mathrm{DKK}$$

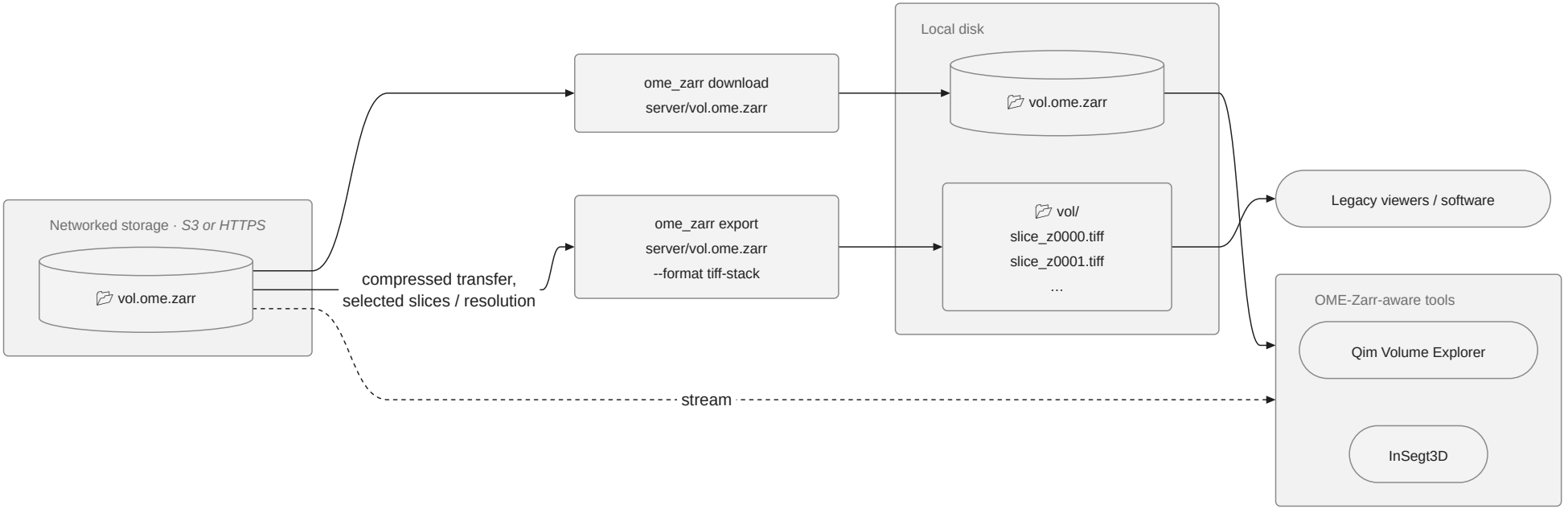
Caveat: for the datasets provided by museum

- Seems to compress better than CIL reconstructed wolf skulls; lossless CR ≈ 2.8!

NaturArv3D Considerations

- Support for most codecs not yet standardised in Zarr (ongoing work)
  - Without standardisation: ecosystem split between Python and Rust implementations
- Most software doesn’t support OME-Zarr yet
  - Workaround: use Zarr streaming + local tool
  - Stream Zarr but split out TIFF stacks locally
    - Compressed during transfer
    - Can access specific slices or resolutions
  - Currently trying to get this into `py-ome-zarr`
    - Open PRs, ongoing work

NaturArv3D Considerations



- Store in modern, compressed format
- Streaming export to TIFF stack (supporting older tools)
- Streaming to modern tools

Future Work

- When first data + infrastructure for scanner becomes available:
  - Pick specific codec and parameters
- Consider evaluating some task-specific metrics
  - Segmentation accuracy, etc
- Similar study is pursued by Human Organ Atlas (HOA)
  - Suraj Pathak (HOA, UCL) attended hackathon last week and discussed this
  - They have similar problems and interests
  - Potentially some future collaboration

Questions?

Bibliography

[1] F. Marone, J. Vogel, and M. Stampanoni, “Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices,” *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

[2] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and R. Herfkens, “Real-Time Compression of Raw Computed Tomography Data: Technology, Architecture, and Benefits,” presented at the SPIE Medical Imaging, E. Samei and J. Hsieh, Eds., Lake Buena Vista, FL, Feb. 2009, p. 72582H. doi: 10.1117/12.810599.

[3] “DICOM PS3.5 2026c — Data Structures and Encoding.” Accessed: Jul. 17, 2026. [Online]. Available: https://www.dicomstandard.org/current

Meta

- Slides and source code available at: TODO
- Study on compression strategies available:
  - github.com/qim-center/consult-xct-compression